

#1. The human hearing system operates in the range of 2 – 20,000Hz. What sampling rate will be sufficient to preserve the information content of the signals in this range?

REZ : Cf. Nyquist theory, the sampling rate must be at least twice the bandwidth. Requested bandwidth is: $20000 - 2 = 19998\text{Hz}$, so the necessary for sampling is $2 \cdot 19998 = 39,996$ samples/sec.

#2. Un semnal vocal opereaza in domeniul de frecvente 0-4000 Hz. Ce rata de esantionare ar fi suficienta pentru transmiterea digitala corecta a acestui semnal? Daca se folosesc 8 biti pe esantion, care este rata de transmisie necesara?

REZ : Cf. Teoriei lui Nyquest, rata de esantionare trebuie sa fie de 2 ori latimea de banda. Latimea de banda este $4000 - 0 = 4000$ Hz, deci rata de esantionare ar fi de $2 \cdot 4000\text{Hz} = 8000$ esantionari/secunda. $8 \text{ biti} \times 8000 \text{ esantionari} = 64\,000 \text{ Biti} = 64 \text{ KB/s}$

#3. Vocea umana are un range de 400-3400Hz. Care e frecventa de esantionare necesara? Care e BW uzual pe liniile telefonice? **REZ :** Frecventa = $2 \cdot (3400 - 400) = 6000$ esantionari/secunda. Un semnal de voce (uzual) are bandwidth de 4 kHz si data rate de 64 kbps.

#4. Given a link with a signal/noise ratio of 1023, what bandwidth is required to support the transmission rate from previous problem? At the required bandwidth, how many bits will be transmitted per Hertz?

REZ : Cf. Shannon theorem ($v = H \cdot \log_2(1 + S/N)$); the transmission rate (channel speed) is 13,824,000bps. $H = 13,824,000 / \log_2(1 + 1023) = 13,824,000 / 10 = 1,382,400\text{Hz}$, aprox. 1,3MHz.

#5. Canal, intended capacity = 20 Mbps, bandwidth = 3MHz; S/N = ?

REZ : $20 \cdot 10^6 = 3 \cdot 10^6 \cdot \log_2(1 + S/N) \Rightarrow S/N = 2^{(20/3)} - 1 \text{ dB}$

#6. Signaling system, 9600 bps; encodes a 8-bit word; minimum bandwidth = ?

REZ : Nyquist – $v = 2 \cdot H \cdot \log_2 N \Rightarrow 9600 = H \cdot 16 \Rightarrow H = 600$

#7. Modem cu 1800 baud, codifica pe 4 biti. Data rate=? **REZ :** data rate = baud*enc bits = $1800 \cdot 4 = 7200$ bps

#8. Teleprinter 300 Hz ; S/N = 3 dB v=? **REZ :** $10 \cdot \log_{10} S/N = 3 \rightarrow S/N = 1.9 \rightarrow v = 300 \cdot \log_2 2.9$ bps

#9. a) Telex network; bandwidth = 500Hz ; S/N = 5dB ; b) switched telephone network; bandwidth = 3100 Hz ; S/N = 20dB; Information rates = ? **REZ :** a) $10 \log_{10} S/N = 5 \Rightarrow S/N = 3.16 \Rightarrow v = 500 \cdot \log_2 4.16$; b) $10 \log_{10} S/N = 20 \Rightarrow S/N = 100 \Rightarrow v = 3100 \cdot \log_2 101$

#10. Tv channel; 6Mhz wide; noise = 0 ; 4-level digital signal; data rate = ?

REZ : $v = H \cdot \log_2 N = 6 \cdot 10^6 \cdot \log_2 4 = 12 \text{ Mbps}$.

#11. 3 kHz channel; S/N = 20dB ; max data rate = ? **REZ :** $v = 3 \cdot 10^3 \cdot \log_2 21 = 3 \cdot \log_2 21 \text{ Kbps}$

#12. Why has the PCM sampling time been set at 125microsec? **REZ :** the BW is 62.5 Hz (Nyquist Th.)

#13. In order to transmit an uncompressed video stream at 30 frames/second into a quarter size VGA window (160 * 120 pixels), where each pixel requires 24bits for colour, what transmission capacity is required?

REZ : Total number of pixels in a window: $160 \cdot 120 = 19,200$ pixels. Total number of bits requested within a window: $19,200 \cdot 24 = 460,800$ bits. Number of bits for 30 frames (number of bits sent on a second): $460,800 \cdot 30 = 13,824,000$ bits so there is a need for a transmission speed of approx. 13.8Mbps.

#14. a) Source $480 \cdot 500$ pixels; pixel has 32 intensity values; 30 pictures per second; Data Rate = ? ; b) channel with BW = 4.5 MHz and S/N = 35 dB; capacity = ? **REZ :** a) 32 de valori $\Rightarrow 5$ biti/pixel $\Rightarrow 30 \cdot 5 \cdot 480 \cdot 500 = 36$ Mbps ; b) $10 \log_{10} S/N = 35 \Rightarrow S/N = 10^{3.5}$; $v = 4.5 \cdot 10^6 \cdot \log_2(1 + 10^{3.5})$ bps ;

#15. Vrei video real, ai 160×120 pixeli fereastră, vrei 20fps si ai 127 culori pe pixel. Ce BW e necesar? Ce bw e necesar dacă transmite pe o linie cu raport semnal/zgomot de 1023?

REZ : Nr total de pixeli dintr-o fereastră $160 \times 120 = 19200$ pixeli ; Numarul de biti transmiși într-o fereastră = $127 \cdot 19200 = 2.438.400$; Numarul de biti pentru 20 de frame-uri : $2.438.400 \cdot 20 = 48.768.000$; Cf Shannon ($v = H \cdot \log_2(1 + S/N)$), H ar fi $H = 48.768.000 / \log_2(1 + 1023) = 48.768.000 / 10 = 4.876.800$ Hz = 4,87 KHz

#16. A full duplex 64,000bps point-to-point data link was observed for 60 sec. During the observation period, the following were obtained: 50 original data packets, each containing 24 header bytes and 1000 data bytes; 5 additional data bytes observed to be retransmissions; 100 acknowledgements , each containing 24 header bytes and no data; 4 connection management packets, each containing 124 bytes. What was the channel utilization?

REZ : The bytes sent with data packets: $50 \cdot (24 + 1000) = 51,200$ bytes. The bytes sent for retransmission: 5 bytes. The bytes for acknowledgements: $100 \cdot 24 = 2400$ bytes. The bytes for connection management: $4 \cdot 124 = 496$ bytes. The total of sent bytes: $51,200 + 5 + 2,400 + 496 = 54,101$ bytes. 1 byte = 8 bits $\Rightarrow$ a number of: $54101 \cdot 8 = 432,808$ bits. Theoretically during 60 sec, channel could carry: $64,000 \cdot 60 = 3,840,000$ bits. The utilisation of the channel is: number of sent bits / theoretical number = $432,808 / 3,840,000 \sim 9\%$.

#17. The velocity of propagation of an electrical signal = 70% of the speed of the light. Determine the ratio of the signal propagation delay to the transmission delay, for the following types of data link and 1000 bits of data: a). 100m of UTP wire and a transmission rate of 1Mbps; b). 0.5km of coaxial cable and a transmission rate of 10Mbps; If the signal propagates with the speed of the light, the same question for : c). A satellite link and a transmission rate of 512Kbps; d). 2.5km of fiber optic and a transmission rate of 1000Mbps

REZ : a) transmission delay = N/v ; propagation delay = $d/\text{propagation speed}$; prop speed = $70/100 \cdot 3 \cdot 10^8 = 21 \cdot 10^7$ m/s ; prop delay = $100/(21 \cdot 10^7) = 4 \cdot 10^{-7}$; trans delay = $10^3/10^6 = 10^{-3}$ bps; ratio = $1/(21 \cdot 10^5) \cdot 10^3 = 4 \cdot 10^{-4}$; b) t.d. = $100/10 \cdot 10^6 = 10^{-5}$; p.d. = $5 \cdot 100/21 \cdot 10^7$; ratio = $10^{-5} \cdot 21 \cdot 10^5/5 = 4.2$; c) t.d. = $10^3/512 \cdot 10^3$; p.d. = $d/21 \cdot 10^7$; ratio = $t.d. \cdot 1/p.d. = 4 \cdot 10^5 \cdot d$; d) t.d. = $10^3/10^3 \cdot 10^6 = 10^{-6}$; p.d. = $25 \cdot 10^2 / 21 \cdot 10^5$; ratio = $t.d. \cdot 1/p.d. = 84 \cdot 10^{-5}$;

#18. Consider the use of 1000 bit frames on a 1Mbps satellite channel with a 270ms delay. What is the maximum link utilization for: Stop-and-wait protocol? Continuous flow control with a window size of 7? But with a window of 127? What about a window of 255?

REZ : Bit time: $1/10^6 = 10^{-6}$ sec = 1μsec. Frame time: $1,000 \cdot 1 \mu\text{sec} = 1$ ms. Assuming no extra delay at the receiver, one bit echo will take 270ms for being received by the sender. (Stop-and-

wait): Total transmission time for a frame is: 1 (frame time) + 270 (receiving ACK delay) = 271ms. So effective utilization of the channel is $1/271 = .3\%$. (Window of 7): Sender can send 7 messages (frames) in advance, without ACK, until an enforced wait. 7 frames transmitted in 7ms. There is a 270ms delay after first frame, for the ACK (or RR) of the first frame. Each ACK for following frames takes other 1ms. Thus 7 frames sent correctly in: $7 + 270 = 277\text{ms}$, so an utilization of : $7/277 = 2.5\%$. (Window of 127): 127 frames transmitted every 397ms ($127 + 270$); utilization of: $127/397 = 40\%$. (Window of 255): $255/(255+270) = 51\%$.

#19. LAN CSMA/CD are viteza de transfer 10Mbps si lungimea 1km. Viteza de propagare este de 200.00 km/s. Cadrele de date au o lungime de 256 bti incluzind 32 de biti de antet si suma de control. Cadrul de confirmare e de 32 biti. Care este viteza efectiva? **REZ :** $R = 10 \text{ Mbps}$, $L = 1000 \text{ m}$, $v = 200 \text{ m}/\mu\text{s}$, $D = 256 \text{ biti}$

$T = T_1 + T_2 + \text{intirzierea_de_1_bit}$; $T_1 = \text{timpul_inainte} = t_{\text{de_punere_pe_linie}} + t_{\text{transport_al_ultimului_bit}}$; $T_2 = \text{timpul_inapoi} = t_{\text{de_punere_pe_linie}} + t_{\text{transport_al_ultimului_bit}}$; $T_1 = D/R + L/v = 256/10 \cdot 10^6 + 5\mu\text{s} = 30.6 \mu\text{s}$; $T_2 = 32/R + L/v = 32/10 \cdot 10^6 + 5\mu\text{s} = 8.2\mu\text{s}$; $T_{\text{bit}} = 1/R = 0.1\mu\text{s}$; $T = 30.6 + 8.2 + 0.1 = 38.9\mu\text{s}$; $\text{nr_de_biti_transmisi-eactiv} = D - 32 = 224 \text{ biti}$; $V_{\text{efectiva}} = 224/38.9 = 5.75 \text{ Mbps}$.

#20. Considering building a CSMA/CD network running at 1Gbps over a 1km cable with no repeaters. The signal speed is 200,000km/s. What is the minimum frame size?

REZ : Network transmission time: $1 / 10^9 = 10^{-9} \text{ s}$. For detecting eventual collision, frame transmission must last at least double the propagation time within that medium (round trip delay). Propagation time: $1 / 200,000 = 0.5 \cdot 10^{-5} \text{ s}$, so its double will be 10^{-5} s . Number of bits in a minimum valid frame: $10^{-5} / 10^{-9} = 10,000 \text{ bits}$.

#21. Considering a Token Ring network at a bit rate of 4Mbps, and a signal propagation speed of 200m/ μsec , and each interfaces introducing a delay of one bit period, find the cable length equivalent with a total of 20 interfaces on that ring.

REZ : One's interface delay = One bit period: $1/(4 \cdot 10^6) = 0.25 \cdot 10^{-6} \text{ sec} = 250 \text{ nsec}$. 20 interfaces delay: $20 \cdot 250 \text{ nsec} = 5\mu\text{sec}$. Propagation speed: 200m/ μsec , so for 5 μsec the cable equivalence is 1,000m.

#22. Considerati o retea Token Ring la o rata de transmisie de 16Mbps, si o viteza de propagare a semnalului de 200m/microsec, fiecare interfata atasata introducand o intarziere de o perioada de bit. Gasiti lungimea de cablu echivalenta ce introduce aceeasi intarziere ca cea introdusa de o interfata. Ce efect ar avea un numar prea mare de interfete active atasate la inel ?

REZ : Fiecare interfata are un delay de : un bit $\Rightarrow 1/(16 \cdot 10^6) = 0.0625 \cdot 10^{-6} = 62.5 \text{ nsec}$. Viteza de propagare este de 200m/microsec, astfel pentru 62.5nsec = 0.0625 microsecunde, distanta are fi : 12.5 metri. Un numar prea mare de interfete ar avea ca efect cresterea timpului de propagare a token-ului si deci a comunicarii in retea.

#23. Token Ring la 10Mbps, lung de 1 km, viteza de propagare de 200m/ μs , 15 statii uniform raspindite, cadrele de date sint de 256 biti incluzind 32 de biti de incarcare suplimentara., confirmarile sint atasate la cadrele de date si sint gratuite, jetonul e de 8 biti. Care este viteza efectiva? **REZ :** $T = T_1 + T_2$; $T_1 = \text{timpul de parcurs al inelului de catre token}$; $T_2 = \text{timpul de parcurs al inelului de catre packet}$; Apare intirzierea de 1 bit introdusa de fiecare statie si intirzierea de propagare a semnalului (la token); $T_1 = \text{intirzierea_de_1_bit} + \text{timp_punere_jeton} + \text{timp_statii} = L/v + \text{dim_jeton} \cdot 1/R + \text{dim_jeton} \cdot \text{nr_statii} \cdot 0.1\mu\text{s} = 5 \mu\text{s} + 8 \cdot 0.1\mu\text{s} + 8 \cdot 15 \cdot 0.1\mu\text{s} = 17.8 \mu\text{s}$; $T_2 = \text{timpul_de_parcurs_al_unui_bit} + \text{timpul_de_punere_pe_linie} = L/v + D/R = 1000\text{m}/200 \text{ m}/\mu\text{s} + 256/10\text{Mbps} = 5\mu\text{s} + 25.6\mu\text{s} = 30.6\mu\text{s}$; $T = 30.6 + 17.8 = 48.4 \mu\text{s}$; $V_{\text{efectiva}} = 224/48.4 = 4.62 \text{ Mbps}$.

#24. Token Ring cu $R = 5 \text{ Mbps}$, $v = 200 \text{ m}/\mu\text{s}$. Cu citi metri de cablu e echivalenta o intirziere e 1 bit?

REZ : $R = 5 \text{ Mbps} \Rightarrow t_{\text{bit}} = 1/R = 0.2 \mu\text{s}$; $D = v \cdot t = 200 \text{ m}/\mu\text{s} \cdot 0.2 \mu\text{s} = 40 \text{ m}$ cablu.

#25. Care este adresa subnet, broadcast, si intervalul de adrese valide cunoscind o adresa IP din retea 172.16.10.5 si mask-ul 255.255.255.128 ? **REZ :** 172.16.10.5 – adresa de clasa B ; clasa B : Network pe 16 biti, Host pe 16 biti. $255.255.255.128 = 255.255.11111111.10000000 \Rightarrow 9$ biti pentru codare subnet, 7 biti pentru codare host-uri (ma uit la ultimii 16 biti, tot ce e cu 1 e pe partea de subnet, ce e cu 0 pe partea de host); $172.16.10.5 = 172.16.00001010.00000101$; calculare adresa subnet - metoda 1 : copiez adresa IP care mi se da in ipoteza, si in partea de host pun 0-uri (aici ultimii 7 biti) $172.16.00001010.00000000 = 172.16.10.0$; metoda 2 : SI logic intre adresa IP pe care o am si mask; $172.16.00001010.00000101 \wedge 255.255.11111111.10000000 \Rightarrow 172.16.00001010.00000000 = 172.16.10.0$; calculare adresa broadcast : se obtine din adresa IP data care va avea in partea de host numai 1; adresa data: $172.16.10.5 = 172.16.00001010.00000101$; adresa broadcast: $172.16.10.01111111 = 172.16.10.127$; calculare intervala de adrese valide: pentru codificarea host-urilor se utilizeaza 7 biti $\Rightarrow$ se pot aloca $2^7 - 2 = 126$ de adrese in intervalul 172.16.10.1 – 172.16.10.126.

#26. Cerintele de la #26, dar adresa 192.168.100.17 si se utilizeaza 4 biti pentru codificare subretele.

REZ : 192.168.100.17 – adresa de clasa C; clasa C : network pe 24 biti, host pe 8 biti. mask-ul implicit pentru clasa C este 255.255.255.0, dar in ipoteza se specifica faptul ca se utilizeaza 4 biti pentru subnet (vor fi 4 biti din ultimul octet al adresei de clasa C, biti codati cu 1) $\Rightarrow$ mask-ul utilizat va fi 255.255.255.11110000 = 255.255.255.240; calculare adresa subnet: $192.168.100.17 = 192.168.100.00010001$ (adresa de clasa C, nu pot sa lucrez decit in ultimul octet), adresa subnet-ului : $192.168.100.00010000 = 192.168.100.16$ (in partea de host, ultimii 4 biti din ultimul octet “umplu” cu 0) ; calculare adresa broadcast: $192.168.100.00011111 = 192.168.100.31$ (bitii de host sint 1); calculare interval de adrese: 192.168.100.17 – 192.168.100.30.

#27. Se da adresa de retea 192.168.100.0 si mask-ul 255.255.255.224. **REZ :** a) de ce clasa este adresa si citi biti din partea de host au fost imprumutati pentru subnetting? 192.168.100.0 – adresa de clasa C (8 biti pe partea de Host) ; $255.255.255.224 = 255.255.255.11100000 \Rightarrow 3$ biti pt subnetting, 5 biti pentru host-uri ;

b) cite subretele pot construi si cite host-uri pot avea intr-o subretea, numarul total de host-uri din retea? 3 biti pentru subretele $\Rightarrow 2^3 - 2 = 6$ subretele; 5 biti pentru host-uri $\Rightarrow 2^5 - 2 = 30$ host-uri/subretea; numarul total de host-uri din retea = $6 \cdot 30 = 180$ host-uri; c) care este adresa celei de a 4-a subretele? $192.168.100.10000000 = 192.168.100.128$; d) care este adresa de broadcast a celei de a 2-a subretele? adresa subretea 2 = $192.168.100.01000000 = 192.168.100.64$; adresa de broadcast (bitii de host ii fac 1) $\Rightarrow 192.168.100.01011111 = 192.168.100.95$; e) din a cita subretea face parte adresa 192.168.100.188 ; $192.168.100.188 = 192.168.100.10111100 \Rightarrow$ a 5-a subretea (101).

#28. Folosind notatia zecimala cu punct, adresa IP a unui calculator este: 150.7.26.123, iar masca de subretea este : 255.255.255.240. a) Din ce clasa de adrese face parte adresa de mai sus si care este adresa de retea din care face parte calculatorul? b) Care este structura acestei adrese dupa aplicarea mastii de retea? Cati biti are fiecare camp din structura adresei? c) Daca acest calculator trebuie sa transmita un pachet de broadcast, care este adresa IP destinatie?

REZ : a) IP-ul este de clasa B, iar adresa de retea este 150.7.26.112; b) 4 biti pentru host, 28 de biti pentru adresa subretelei. ; c) 150.7.26.137

#29. IP: 129.6.31.101 subnet mask: 255.255.255. 192. a) what class is IP; b) after applying subnet mask, what is the structure of the address. How many bits are used for net, subnet, host identifiers? ; c) If host desires to broadcast to all hosts on the same network and subnet as itself, then what destination is chosen?

REZ : a) Clasa B (cuprins intre 128.1.0.1 - 191.255.255.254) ; b) 129.6.31.64 (logic AND intre IP si SubnetMask) ; c) 129.6.31.127 (logic AND intre IP si SubnetMask, apoi bitii de host ii punem 1).